- Announcements:
  - Ass 2 due today @ 4pm
  - Ass 3 release on Monday

Recap:

1) A pivot is a theoretical (& non-computable) transformation of a random variable whose distn no longer depends on any unknown parameters
2) Our level of confidence in any estimation procedure must always be tied to some margin of error
3) Margins of error can be additive, or multiplicative (or...)

Confidence intervals for pop^n proportion p

In general, pivot transforms for discrete r.vs do not exist. However, we can still construct asymptotic pivots.

Eg 1 (Skittles revisited). From a sample of n with, we observe X successes, where we don't know the probability of success, p. A model for X would be

$$X \sim Bin(n, p)$$ where p is unknown.

Recall, the sample proportion $\hat{p} = \frac{X}{n}$ has expected value p (1)

& its standard error $se(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ (2)

& (3) we also know from CLT that $\hat{p}$ has an asymptotic normal distn.

Hence, combine (1), (2) & (3) gives us the transformation.

$$\frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \longrightarrow N(0,1) \text{ for "large } n\text{"}$$

which is an asymptotic pivot.

Thus, for an approx 95% CI, say, for the pop^n proportn p, we can again start with a fully known prob. statement:

$$95\% = P(-1.96 \leq N(0,1) \leq +1.96)$$

$$= P\left(-1.96 \leq \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \leq +1.96\right)$$

rearrange this to give us an approx 95% CI:

$$\hat{p} \pm 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \quad \text{for large } n$$

This method is only approximate, because the CLT requires n to be "increasingly large". So, how large is large enough for n?

A general rule of thumb is that the sample size n has to be large enough to observe at least 10 successes & at least 10 failures.

If n is not large enough, we can use alternative methods (via simulations, invert hypothesis tests, or...)

General defn of a confidence interval procedure

Let $X_1, X_2, \ldots, X_n$ be data coming from a pop^n which depends on some parameter θ. Then

$\left(T_1(\underline{X}), T_2(\underline{X})\right)$ is an exact $(1-\alpha)100\%$ CI for θ if

$T_1(\underline{X}), T_2(\underline{X})$ are functions of data alone

& $P\left(T_1(\underline{X}) \leq \theta \leq T_2(\underline{X})\right) = 1-\alpha$ $\forall\, \theta \in \Theta$

↑ fixed

↖ ↗ random endpoints that only depend on the data

If $P\left(T_1(\underline{X}) \leq \theta \leq T_2(\underline{X})\right) \geq 1-\alpha$ $\forall\, \theta \in \Theta$,

then this called a conservative CI with level (at least) $1-\alpha$.

Notes: (a) interval estimates are more useful than a single point estimate, because they provide a measure of uncertainty about your estimate along with a level of confidence. This is the key difference between Statistics & ML.

(b) Confidence interval procedures for comparing two samples - see reference notes, tutorials & assignments.

(c) CIs can be considered as a range/set of plausible parameter values that are consistent with the observed data. In this sense, CIs can be considered as the inverse of hypothesis testing (week 8)

(d) The probability statement that we use for CI's tells us how often the procedure works before taking our sample / carrying out our experiment. We must be careful to use the word "confidence" once we have collected our data & enumerated our confidence interval.

Can we ever to use the word "chance" after our experiment is carried out? Eg. After taking your first exam, can you ever say "There's a 60% chance I passed?"

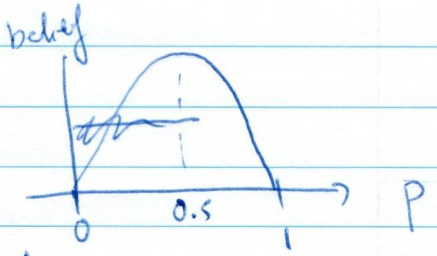
Yes, if you take a Bayesian interpretation of probability.

Bayesian uncertainty & credible intervals

Before any skittles & M&M's were sampled, what was your prior belief about the "p" proportion of skittles?

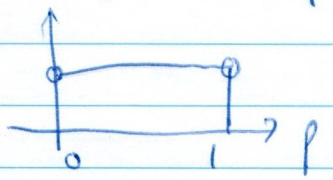
Max: "around 50-50"

$$f(p) \propto p^{2-1}(1-p)^{2-1}$$



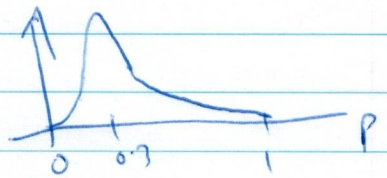
Jack: "no opinion"

$$f(p) = 1 \quad 0 < p < 1$$
$$= p^{1-1}(1-p)^{1-1}$$



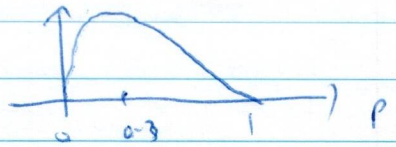
Alan: "around 30-70"

$$f(p) \propto p^{30-1}(1-p)^{70-1}$$



Other: "around 30-70"

$$f(p) \propto p^{3-1}(1-p)^{7-1}$$



These are all examples of a Beta(α, β)

After collecting our data, we observed $x = 23$ successes out of $n = 47$ trials. So our likelihood is

$$f(x = 23 \mid p) \propto p^{23} (1-p)^{24}$$

[In general $f(x \mid p) \propto p^x (1-p)^{n-x}$]

We can now update our belief after seeing the data using Bayes' rule:

$$f(p \mid x) = \frac{f(p, x)}{f(x)}$$

posterior belief

$$= \frac{f(x \mid p) f(p)}{f(x)} \quad \text{Bayes' rule.}$$

$$\propto f(x \mid p) f(p)$$

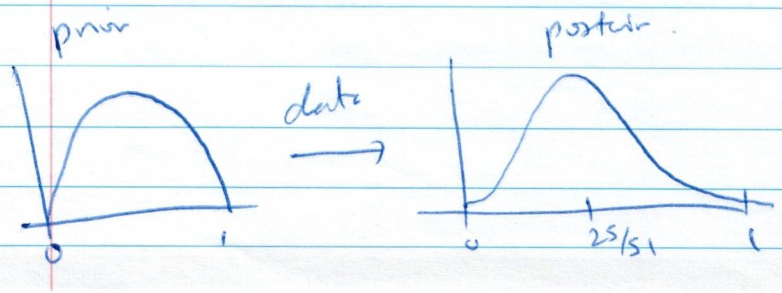
likelihood, prior belief

For Max: prior belief was $f(p) = p^{2-1} (1-p)^{2-1}$, but after seeing the data of $x = 23$ out of $n = 47$, posterior belief is updated to

$$f(p \mid x = 23) \propto \overbrace{p^{2-1} (1-p)^{2-1}}^{\text{prior}} \overbrace{p^{23} (1-p)^{24}}^{\text{likelihood}}$$

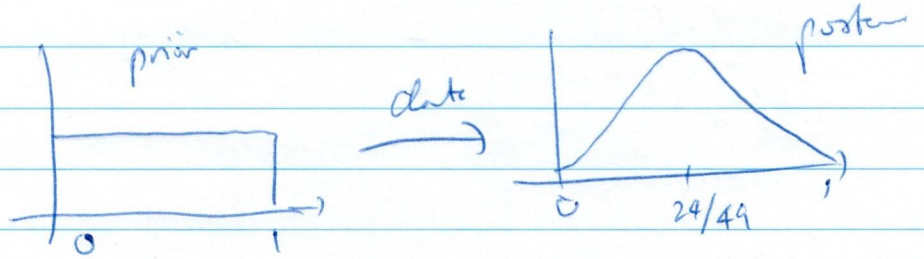
$$= p^{25-1} (1-p)^{26-1}$$

$$\sim \text{Beta}(25, 26)$$



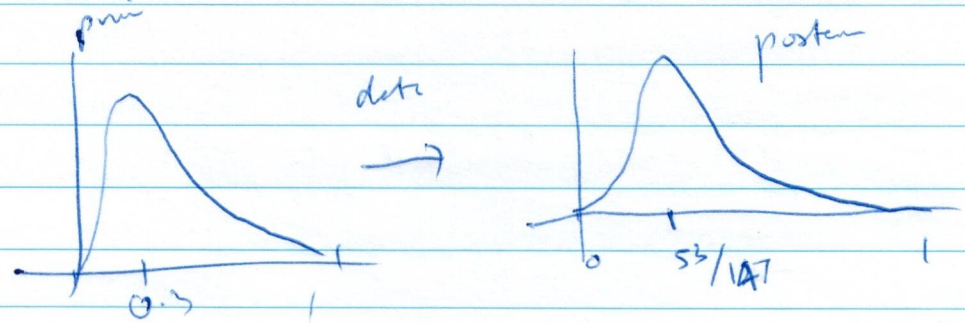
Jack: $f(p) \propto 1$, so posterior belief is

$$f(p \mid x=23) \propto f(p)\, f(x \mid p)$$
$$= p^{1-1}(1-p)^{1-1}\, p^{23}(1-p)^{24}$$
$$= p^{24-1}(1-p)^{25-1}$$
$$\sim \text{Beta}(24, 25)$$



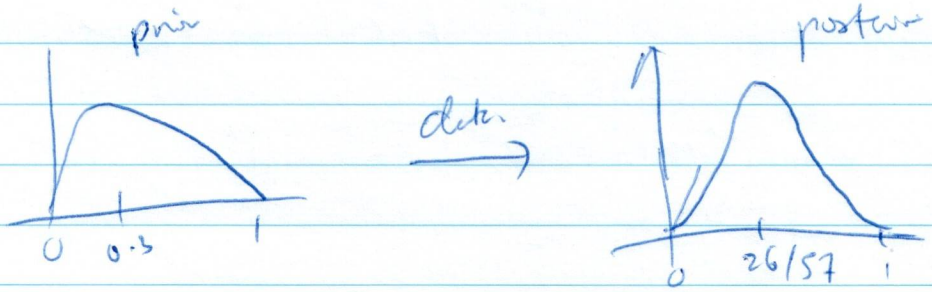
Alan: $f(p) \propto p^{30-1}(1-p)^{70-1}$, so posterior

$$f(p \mid x=23) \propto p^{30-1}(1-p)^{70-1}\, p^{23}(1-p)^{24}$$
$$= p^{53-1}(1-p)^{94-1}$$
$$\sim \text{Beta}(53, 94)$$



Other: $f(p) \propto p^{3-1}(1-p)^{71}$, so posterior

$$f(p \mid x=23) \propto p^{3-1}(1-p)^{71}\, p^{23}(1-p)^{24}$$
$$= p^{26-1}(1-p)^{31-1} \sim \text{Beta}(26, 31)$$



A point of controversy is that different prior beliefs lead to different posterior beliefs even after seeing the same dataset.

However, this is OK if we can interpret probabilities as a personal level of belief about something.
For example, after complete the same final exam, your own belief about passing may be different to another student's belief about passing.

→ Credible intervals next week!
→ posterior mean, mode.